

YEAR 7 • UNIT 1 • HOMEWORK 5

Unit 1, All of It

Science 1.1–1.4 — cells, specialised cells, tissues and organs

Mathematics 1.1–1.6 — integers, factors, multiples, tests and roots

Name: _____

Class: _____

Date given: _____

Date due: _____

How to do this homework

This packet is **the whole of Unit 1**, in one place. Science is at the front this time; the maths is longer, so it is at the back.

- In maths, **write the calculation before the answer**. Every time.
- In science, answer in **full English sentences**. “Organ” is not a sentence; “The heart is an organ” is.

Section A • Science 1.1–1.4 — Cells, Tissues and Organs**A1 Match the word to its meaning**Write the correct **letter** in each box.

Word		Meaning
1. Organelle	☐	A A group of organs working together, like the heart and the blood vessels.
2. Tissue	☐	B The strong material a plant cell wall is made of.
3. Organ	☐	C A whole living thing, like a frog or an oak tree.
4. Organ system	☐	D A part inside a cell that does one job.
5. Organism	☐	E Changed in shape so it can do one job very well.
6. Chlorophyll	☐	F A group of similar cells working together, like muscle.
7. Specialised	☐	G The green substance inside a chloroplast that traps light.
8. Cellulose	☐	H Several tissues working together, like the heart or a leaf.

A2 Climb the ladder

Word help – the ladder, in order

cell → **tissue** → **organ** → **organ system** → **organism** — and each rung is **made of** the rung below it. A tissue is made of cells; an organ is made of tissues.

For each thing, write which **level** it is, and then what the **level above it** would be. One of these has no level above it — think carefully before you write.

Thing	Which level?	The level above it
A red blood cell		
Muscle		
The stomach		
The digestive system		
Mr Bowen		

A3 Name the cell

Remember from the lesson

A cell's **function** is the job it does. A **specialised** cell has a structure that helps it do that job well. The sentence to use: A [cell] is **adapted to** [its job] **because it has** [its feature].

(a) Each clue describes one specialised cell. Name it.

(i) It has no nucleus, which leaves more room for the oxygen it carries. _____

(ii) It is very long and thin, so a message travels a long way along it. _____

(iii) It has tiny hairs along the top that sweep dust out of your lungs. _____

(iv) It has a long thin part that reaches into the soil for water. _____

(v) It is packed with chloroplasts and sits near the top of a leaf. _____

(b) Now write the full sentence for **two** of them, using **adapted to** and **because it has**.

(c) A palisade cell is full of chloroplasts. A root hair cell has **none at all**. Suggest why not.

A4 Answer in full sentences**Sentence starters – you may copy these**

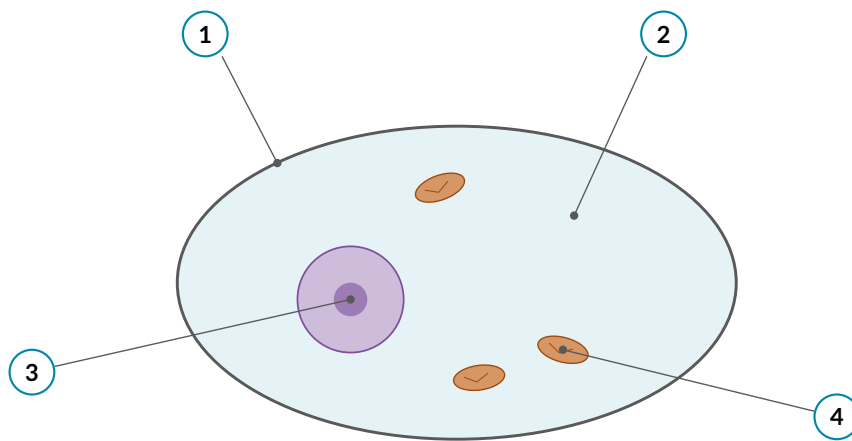
“I know it is a ... cell because...”

“It also has..., and an animal cell has no...”

Mr Bowen looks at one cell under the microscope. It has a straight, stiff edge, and one large space fills most of the middle of it. Is it a **plant** cell or an **animal** cell? Say **how you know**, in full sentences.

A5 Label the animal cell

Write the name of each numbered part on the lines below the diagram.



1. _____

3. _____

2. _____

4. _____

Now think. Name **two** things a **plant** cell would have that this cell does not.

Section B · Mathematics 1.1–1.2 – Integers

The rules are on the next page. Try each question **before** you look at them – if you can, you know them.

The two sets of rules, side by side

Add a positive · move **right** $(+) \times (+) = (+)$

Add a negative · move **left** $(-) \times (-) = (+)$

Subtract a positive · move **left** $(+) \times (-) = (-)$

Subtract a negative · move **right** $(-) \times (+) = (-)$

Two negatives make a positive is true for $\times$ and $\div$. It is **not** true for $+$: $-6 + (-4) = -10$, not -2 .

B1 Higher, lower, warmer, colder

Complete each sentence with **higher than**, **lower than**, **warmer than** or **colder than**.

(i) -9 is _____ -2 .

(ii) A temperature of -1°C is _____ a temperature of 6°C .

(iii) Floor 3 is _____ floor -1 .

B2 All four rules, and all four signs

Work these out. You may draw a small number line beside any of them.

(a) $-7 + 3 =$ _____

(e) $-8 \times 3 =$ _____

(i) $-20 + (-14) =$ _____

(b) $6 + (-11) =$ _____

(f) $-6 \times (-7) =$ _____

(j) $12 - (-12) =$ _____

(c) $-4 - 9 =$ _____

(g) $45 \div (-9) =$ _____

(k) $-4 \times (-5) \times (-2) =$ _____

(d) $-5 - (-12) =$ _____

(h) $-72 \div (-8) =$ _____

(l) $-30 \div 6 \times (-2) =$ _____

B3 Say it, then write it

Word help – these words decide the order of your calculation

subtract 5 from 8 you **start** at 8: $8 - 5$. The English gives the numbers in the **opposite order** to the calculation.

take 6 away from 2 the same trap: $2 - 6$.

7 less than 3 again the same trap: $3 - 7$.

owe / a debt money you must pay back. It makes your total **negative**.

pay back to give owed money **back**, so the debt gets **smaller**.

Turn each English sentence into a calculation, then work it out. **Write the calculation first.**

1. Subtract 11 from 5.

Calculation: _____ Answer: _____

2. Take 8 away from -3 .

Calculation: _____ Answer: _____

3. What is 9 less than 2?

Calculation: _____ Answer: _____

4. Mr Bowen owes 45 dollars. He pays back 20 dollars.

Calculation: _____ Answer: _____

B4 Difference

Word help – difference

The **difference** between two numbers is the **gap** between them, so it is never negative – and **the difference between a and b** is **not** the same instruction as **a minus b** . Question 2 is about exactly that.

1. What is the difference between 12 and -5 ? _____

2. (a) The difference between -7 and 4 is _____ (b) $-7 - 4 =$ _____

(c) These two answers are not the same. Explain why.

Section C · Mathematics 1.3–1.6 – Factors, Multiples, Tests and Roots

C1 Build the HCF and the LCM from the lists

One list stops, the other never does

Factors divide into the number. That list **stops**, so we ask for the **highest** common one – the **HCF**.

Multiples are its times table. That list **never stops**, so we ask for the **lowest** common one – the **LCM**.

- (a) All the factors of 48: _____

- (b) All the factors of 60: _____

- (c) The common factors: _____ so the **HCF** is _____

- (d) The first six multiples of 8: _____

The first six multiples of 12: _____

So the **LCM** of 8 and 12 is _____

(e) A student says the LCM of 8 and 12 is 96, because $8 \times 12 = 96$. Is 96 a common multiple of 8 and 12? Is it the **lowest** one? Explain.

C2 Which one does the question want?

Word help – the words that tell you which tool to reach for

together again · **at the same time** · **how often** – something **repeating**. You want the **LCM**.

share into equal groups · **as large as possible** · **none left over** – something being **split up**. You want the **HCF**.

Tick **HCF** or **LCM** for each one, then work out the answer.

The question	HCF	LCM	Answer
One bell rings every 6 minutes, another every 10. They ring together now. In how many minutes will they ring together again?			
Mr Bowen has 24 pens and 36 pencils. He makes identical packs, using everything, with none left over. What is the greatest number of packs?			
Ribbons 45 cm and 60 cm long are cut into equal pieces, as long as possible, with none wasted. How long is each piece?			
A red bus leaves every 12 minutes and a blue bus every 18 minutes. They leave together at 7 am. When do they next leave together?			

C3 Tests for divisibility

(a) Put a tick (✓) where the number divides exactly. Use the tests – do **not** do the divisions.

Number	2	3	4	5	6	9
3942						
5280						
7128						
2475						

(b) **Three sentences, one fact.** We know that $48 \div 6 = 8$ exactly. Fill in the three ways of saying so.

(i) 6 is a _____ of 48. (ii) 48 is a _____ of 6.

(iii) 48 is _____ by 6.

C4 Squares, cubes and roots

Rooting is squaring run backwards

$9^2 = 81$, so $\sqrt{81} = 9$. $4^3 = 64$, so $\sqrt[3]{64} = 4$. And a square number can never end in **2, 3, 7 or 8**.

(a) $7^2 =$ _____

(d) $10^3 =$ _____

(g) $\sqrt[3]{27} =$ _____

(b) $11^2 =$ _____

(e) $\sqrt{81} =$ _____

(h) $\sqrt[3]{125} =$ _____

(c) $4^3 =$ _____

(f) $\sqrt{144} =$ _____

(i) $6^2 + 3^3 =$ _____

(j) Two of these **cannot** be square numbers. Ring them, and say how you know without doing any working.

2500 4832 6400 1007

(k) **Consecutive** numbers follow each other, like 5, 6, 7. The consecutive pair whose squares are 64 and 81 is _____

Section D · Mathematics – Word Problems

Every question here is one of the six maths lessons wearing a story. Before you calculate, ask yourself **which lesson this is** – then write the calculation down before the answer.

1. **Sa Pa.** At 5 am the temperature was -4°C . By 1 pm it had risen by 13 degrees. By 9 pm it had fallen by 9 degrees. (a) What was the temperature at 9 pm? (b) What is the difference between the warmest and the coldest temperature of the day?

2. **The lighthouses.** One lighthouse flashes every 8 seconds and another every 12 seconds. They flash together at exactly 9:00:00. At what time do they next flash together?

3. **The pens.** Mr Bowen has 54 red pens and 36 blue pens. He makes identical packs, using every pen, with none left over, and he wants **as many packs as possible**. How many packs, and how many of each colour goes in one pack?

4. **The patio.** (a) A square patio is made of 169 square tiles. How many tiles are there along one side? (b) A large cube is built from 216 small cubes. How many small cubes are there along one edge?

5. **The biscuits.** Mr Bowen buys a bag of 60 round biscuits and a bag of 48 square biscuits. He puts them on plates so that every plate has the same number of round biscuits and the same number of square biscuits, with none left over, using **as many plates as possible**. (a) How many plates are there? (b) How many biscuits are on one plate? (c) Mr Bowen then eats one whole plate. How many biscuits are left?

D2 Now you write the question

The rules for both

A **real** situation – money, temperature, floors, buses, packing. Full sentences, ending in a **question**. At least one of the **signal words** from this packet. And your own question must actually come out at the answer given.

Write a word problem for this calculation.

$$-7 + 12 = 5$$
